

Certified analytical bounds for hypersonic reachable sets

Ethan Knox

Independent Researcher, Champaign, IL

PhD Research Summary Draft

September 29, 2026

Abstract

A method for computing *certified* reachable sets of a full six-degree-of-freedom hypersonic entry vehicle, reaching state dimensions where grid-based dynamic programming cannot, with a guarantee that is derived analytically and verified in exact arithmetic rather than produced by a numerical solver.

Open items

- Item 3(b) - Exact-Arithmetic Computational Pipeline:** The proposal demands a guarantee verified in exact arithmetic rather than produced by a numerical solver. Floating-point inaccuracies inherent to standard simulation tools compromise the analytical certificate. Must implement a custom, exact-arithmetic interval integrator (likely leveraging C/C++ or specialized rational arithmetic libraries). This pipeline must process the hybrid dynamics, manage the boundary-layer bubble transitions, and compute the intersection geometries deterministically to verify the final bounds. 3
- Item 1(a) - Liouville Equation for Differential Inclusions:** The current framework embeds the flight mechanics $\dot{x} = f_q(x, c, t)$ as a linear Liouville constraint on an occupation measure μ over $[0, T] \times \mathcal{X} \times U$. By bounding true non-polynomial aerodynamics within strict intervals, the vector field becomes a differential inclusion $\dot{x} \in \mathcal{F}(x, c, t)$. We must formulate a robust Liouville equation capable of acting on set-valued maps, and rigorously prove that the support of the resulting occupation measure, $\text{supp } \mu_T$, securely outer-approximates all possible trajectories generated by any vector field selected from the interval inclusion. 3
- Item 2(b) - Wrapping Effect Suppression in Boundary-Layer Bubbles:** Boundary-layer bubbles represent discrete atmospheric skip events as compact impulse sets $\mathcal{I}_j$ carried by internal scaling groups. Because our framework fixes time as the independent variable, integrating interval bounds through highly dynamic skip events will induce severe "wrapping effects" (artificial inflation of bounding boxes). Must develop advanced intersection geometries (zonotopes, Taylor models, etc.) integrated into the profile decomposition to prove geometries maintain tight corridor bounds through multiple skip events without the reachable set exploding prematurely. 3
- Item 1(b) - Bahouri-Gérard Under Interval Disturbances:** The framework relies on the Bahouri-Gérard almost-orthogonality argument to separate the multi-parameter scaling groups (atmospheric, attitude, and elastic). Under interval aerodynamics, these groups couple with an adversarial, bounded uncertainty. We must prove the non-interaction theorem for the defect measures while accounting for interval bounds, guaranteeing that the gap between the upper and lower aerodynamic bounds does not cause mass to leak between scaling groups during atmospheric transition, preserving the compact residual bound $O(\sqrt{\epsilon})$ 4
- Item 3(a) - Analytical Derivation of Corridor Constants:** The ultimate guarantee relies on deriving the residual corridor constant $C_{\mathcal{R}}$ and the maximum boundary-layer bubble count $K_{\max}(Q_{\max})$; The number of bubbles is bounded a priori by the integrated heat-load budget. We must then derive explicit, closed-form algebraic expressions for $C_{\mathcal{R}}$ and $K_{\max}$ strictly from the physical corridor bounds, initial conditions, and the interval widths of the aerodynamics, entirely without relying on numerical interior-point solvers. 4
- Item 2(a) - State Space De-Augmentation and Minimization:** Preliminary plan to fit Lasserre/Henrion moment-SOS required us to introduce additional variables (e.g., $y = \sqrt{\rho/\rho_0}$, $\rho = 1/||\vec{r}||$), which were adjoined to force the system into a polynomial format. Superseding this with interval arithmetic natively handles transcendentals without requiring algebraic embeddings. Therefore we must strip away the polynomial adjunctions and collapse the augmented state space back to its minimal physical dimensions. We also need to determine if retaining certain adjunctions (like quaternion kinematics to avoid coordinate singularities) is computationally beneficial for interval evaluation, or if a raw 13-state formulation is optimal for corridor bounds. 5

The problem

The reachable set of an entry vehicle (every state it can be driven to under admissible control) is what safety and threat analyses need bounded. Computing it exactly means solving a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in state dimension and capped in practice near four or five states. A rigid-body entry vehicle carries thirteen states, $x = (r, \theta, \phi, u, v, w, q_0, q_3, \omega_{1:3}) \in \mathbb{R}^{13}$, and twenty-five once ablation recession, integrated heat load and aerothermoelastic deformation are coupled. So the exact grid-HJI method quits before the real problem starts. The standard polynomial barrier with coefficients obtained from an interior-point solver has analytical *form* but inherits the solver’s accuracy, so it is not a certificate. There are then two problems to solve, not one: dimension, and the provenance of the guarantee.

Item 3(b) - Exact-Arithmetic Computational Pipeline: The proposal demands a guarantee verified in exact arithmetic rather than produced by a numerical solver. Floating-point inaccuracies inherent to standard simulation tools compromise the analytical certificate. Must implement a custom, exact-arithmetic interval integrator (likely leveraging C/C++ or specialized rational arithmetic libraries). This pipeline must process the hybrid dynamics, manage the boundary-layer bubble transitions, and compute the intersection geometries deterministically to verify the final bounds.

The approach

Embed the hybrid flight dynamics into a Banach space of occupation measures. The nonlinear flight mechanics $\dot{x} = f_q(x, c, t)$ become *linear*: the Liouville equation is a linear constraint on a measure μ over $[0, T] \times \mathcal{X} \times U$, and $\text{supp } \mu_T$ outer-approximates the reachable set. So far this is the occupation-measure / moment-SOS program of Lasserre, Henrion and collaborators.

Item 1(a) - Liouville Equation for Differential Inclusions: The current framework embeds the flight mechanics $\dot{x} = f_q(x, c, t)$ as a linear Liouville constraint on an occupation measure μ over $[0, T] \times \mathcal{X} \times U$. By bounding true non-polynomial aerodynamics within strict intervals, the vector field becomes a differential inclusion $\dot{x} \in \mathcal{F}(x, c, t)$. We must formulate a robust Liouville equation capable of acting on set-valued maps, and rigorously prove that the support of the resulting occupation measure, $\text{supp } \mu_T$, securely outer-approximates all possible trajectories generated by any vector field selected from the interval inclusion.

The new content is what happens in the thin-atmosphere limit $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$. Rescaled by the phugoid-to-secular time ratio (which, as in the classical Chapman–Loh variables, enters as $\sqrt{\varepsilon_{\text{atm}}}$, not ε_{atm}), the occupation-measure family *is not tight*. Mass concentrates onto the low-dimensional equilibrium-glide slow manifold $\Lambda = L \cos \sigma / [m(g - V^2/r)] = 1$. This is precisely Lions’ concentration alternative, read on the defect measure of the rescaled family.

Rather than restore compactness, we extract the profile decomposition the dichotomy hands us:

- a finite-dimensional *skeleton* $\mathcal{S}$ —the concentration component, the glide manifold as a graph over its retained coordinates;
- one *boundary-layer bubble* per skip event, the analogue of a concentrating profile: a compact impulse set $\mathcal{I}_j$ of dimension $\leq m$, carried by an internal scaling group at each atmospheric pass;

Item 2(b) - Wrapping Effect Suppression in Boundary-Layer Bubbles: Boundary-layer bubbles represent discrete atmospheric skip events as compact impulse sets $\mathcal{I}_j$ carried by internal scaling groups. Because our framework fixes time as the independent variable, integrating interval bounds through highly dynamic skip events will induce severe “wrapping effects” (artificial inflation of bounding boxes). Must develop advanced intersection geometries (zonotopes, Taylor models, etc.) integrated into the profile decomposition to prove geometries maintain tight corridor bounds through multiple skip events without the reachable set exploding prematurely.

- a *compact residual* with a quantitative bound, $O(\sqrt{\varepsilon})$ in the corridor metric.

The number of bubbles is bounded *a priori*, $K \leq K_{\max}(Q_{\max})$, by the integrated heat-load budget (a trajectory can only afford so many skips). The multi-parameter separation of the three scaling groups (atmospheric, attitude, elastic) is the Bahouri–Gérard almost-orthogonality argument, transported to defect measures; it is the one genuinely new obligation and it sits on the critical path.

Item 1(b) - Bahouri–Gérard Under Interval Disturbances: The framework relies on the Bahouri–Gérard almost-orthogonality argument to separate the multi-parameter scaling groups (atmospheric, attitude, and elastic). Under interval aerodynamics, these groups couple with an adversarial, bounded uncertainty. We must prove the non-interaction theorem for the defect measures while accounting for interval bounds, guaranteeing that the gap between the upper and lower aerodynamic bounds does not cause mass to leak between scaling groups during atmospheric transition, preserving the compact residual bound $O(\sqrt{\varepsilon})$.

The target statement is

$$\mathcal{R}_{\text{cert}}(w) \subseteq \underbrace{\left(\underbrace{\Phi_{\mathcal{S}}(\Theta)}_{\text{skeleton}} \oplus \underbrace{\bigoplus_{j=1}^K \mathcal{I}_j}_{\text{bubbles}} \right)}_{\text{residual}},$$

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory, with $C_{\mathcal{R}}$ and $K_{\max}$ computable from the corridor bounds alone.

Item 3(a) - Analytical Derivation of Corridor Constants: The ultimate guarantee relies on deriving the residual corridor constant $C_{\mathcal{R}}$ and the maximum boundary-layer bubble count $K_{\max}(Q_{\max})$; The number of bubbles is bounded a priori by the integrated heat-load budget. We must then derive explicit, closed-form algebraic expressions for $C_{\mathcal{R}}$ and $K_{\max}$ strictly from the physical corridor bounds, initial conditions, and the interval widths of the aerodynamics, entirely without relying on numerical interior-point solvers.

Item 2(a) - State Space De-Augmentation and Minimization: Preliminary plan to fit Lasserre/Henrion moment-SOS required us to introduce additional variables (e.g., $y = \sqrt{\rho/\rho_0}$, $\rho = 1/||\vec{r}||$), which were adjoined to force the system into a polynomial format. Superceding this with interval arithmetic natively handles transcendentals without requiring algebraic embeddings. Therefore we must strip away the polynomial adjunctions and collapse the augmented state space back to its minimal physical dimensions. We also need to determine if retaining certain adjunctions (like quaternion kinematics to avoid coordinate singularities) is computationally beneficial for interval evaluation, or if a raw 13-state formulation is optimal for corridor bounds.

A Coupled Entry Dynamics and Polynomial Embedding

A.1 Coupled 6-DOF Entry Dynamics

The entry vehicle state comprises the coupled translational, rotational, thermal, ablative, and modal subsystems. Position is carried in Cartesian Earth-centred Earth-fixed (ECEF) components and attitude as a quaternion from the ECEF frame to the body frame. No angle is a state, so the dynamics have no coordinate singularities anywhere on the globe; geocentric latitude and longitude are read out from the position after integration.

Frames.

- $\mathcal{E}$: Earth-centred Earth-fixed (ECEF), rotating with the constant planetary rate $\vec{\omega}_E = \omega_E \hat{z}_{\mathcal{E}}$.
- $\mathcal{B}$: body frame, fixed at the instantaneous center of mass.

The attitude quaternion $\vec{q} = (q_0, q_1, q_2, q_3)^\top$ carries $\mathcal{E} \rightarrow \mathcal{B}$, with direction-cosine matrix

$$\mathbf{R} \equiv \mathbf{R}_{B/E}(\vec{q}) = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 + q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix}, \quad (\text{A.1})$$

quadratic in $\vec{q}$.

Position Kinematics. With $\vec{r} = (r_1, r_2, r_3)^\top$ the ECEF position and $\vec{V} = (u, v, w)^\top$ the planet-relative velocity resolved in $\mathcal{B}$, the kinematic equation is:

$$\dot{\vec{r}} = \mathbf{R}^\top \vec{V}. \quad (\text{A.2})$$

Translational Dynamics. Newton's law holds in the inertial frame; transporting the planet-relative velocity through $\mathcal{E}$ into $\mathcal{B}$, with inertial body rates $\vec{\omega} = (p, q, r_b)^\top$, mass m , aerodynamic force $\vec{F}_a$, thrust $\vec{F}_t$, and the planetary rate resolved in the body frame $\vec{\omega}_E^{\mathcal{B}} = \mathbf{R} \omega_E \hat{z}_{\mathcal{E}}$, the translational dynamical equations are:

$$\dot{\vec{V}}|_{\mathcal{B}} = \frac{1}{m}(\vec{F}_a + \vec{F}_t) + \mathbf{R} \vec{g}(\vec{r}) - (\vec{\omega} + \vec{\omega}_E^{\mathcal{B}}) \times \vec{V} - \vec{\omega}_E^{\mathcal{B}} \times (\vec{\omega}_E^{\mathcal{B}} \times \mathbf{R} \vec{r}). \quad (\text{A.3})$$

Gravity, including the J_2 oblateness term, in ECEF components:

$$\vec{g}(\vec{r}) = -\frac{\mu}{||\vec{r}||^3} \left[\left(1 + \frac{3}{2} J_2 \frac{R_e^2}{||\vec{r}||^2} \left(1 - 5 \frac{r_3^2}{||\vec{r}||^2} \right) \right) \vec{r} + 3 J_2 \frac{R_e^2}{||\vec{r}||^2} r_3 \hat{z}_{\mathcal{E}} \right]. \quad (\text{A.4})$$

Remark A.1. This is the gradient of the J_2 potential $\frac{\mu}{||\vec{r}||} [1 - \frac{1}{2} J_2 (R_e/||\vec{r}||)^2 (3r_3^2/||\vec{r}||^2 - 1)]$; its radial and northward components reproduce the spherical form, with $r_3/||\vec{r}|| = \sin \phi$.

Rotational Dynamics / Euler's Equations. With angular momentum $\vec{h} = \mathbf{I}\vec{\omega}$, combined aerodynamic / reaction control system (RCS) moments $\vec{M} = \vec{M}_a + \vec{M}_{\text{rcs}}$, and the inertia tensor $\mathbf{I} = \mathbf{I}(s, \eta)$ depending on recession depth s and modal amplitudes η (so that $\dot{\mathbf{I}}\vec{\omega}$ is the variable-inertia term), the rotational dynamical equations are:

$$\mathbf{I}\dot{\vec{\omega}} = \vec{M} - \vec{\omega} \times (\mathbf{I}\vec{\omega}) - \dot{\mathbf{I}}\vec{\omega}. \quad (\text{A.5})$$

Quaternion Kinematics. The quaternion is driven by the body rate *relative to* $\mathcal{E}$, $\vec{\omega}_{B/E} = \vec{\omega} - \vec{\omega}_E^{\mathcal{B}}$. For $\vec{a} = (a_1, a_2, a_3)^\top$,

$$\dot{\vec{q}} = \frac{1}{2} \Omega(\vec{\omega} - \mathbf{R} \omega_E \hat{z}_{\mathcal{E}}) \vec{q}, \quad \Omega(\vec{a}) = \begin{pmatrix} 0 & -a_1 & -a_2 & -a_3 \\ a_1 & 0 & a_3 & -a_2 \\ a_2 & -a_3 & 0 & a_1 \\ a_3 & a_2 & -a_1 & 0 \end{pmatrix}. \quad (\text{A.6})$$

Remark A.2. The rate is **polynomial** — bilinear in $(\vec{\omega}, \vec{q})$ and cubic in $\vec{q}$ through the Earth-rate term. With this Ω and $\mathbf{R}$ of Eq. (A.1), $\dot{\mathbf{R}} = -[\vec{\omega}_{B/E} \times] \mathbf{R}$ holds identically, and $\Omega(\vec{a})$ is skew for every $\vec{a}$, so $|\vec{q}|$ is preserved.

Thermal. With the density-square-root embedding $y = \sqrt{\rho/\rho_0}$ and $V = \sqrt{u^2 + v^2 + w^2}$, the modulus lift $\dot{\chi} = c \chi \dot{T}_w$ replaces T_w with a polynomial via Varma–Savageau–Voit recasting. Stagnation-point heating (Sutton–Graves, with nose radius R_n , sea-level density ρ_0) equations are then given by:

$$\dot{Q}_{\text{tot}} = \dot{Q} = k_Q \sqrt{\rho_0/R_n} y V^3, \quad \dot{T}_w = \frac{\dot{Q} - (T_w - T_\infty)/\tau_{\text{cond}}}{C_w}. \quad (\text{A.7})$$

Ablation. With recession rate coupled to heat flux and mass loss from char-layer blow-off, the ablative equations are given by:

$$\dot{s} = \alpha_s \dot{Q}, \quad \dot{m} = -\dot{m}_{\text{abl}}. \quad (\text{A.8})$$

Modal/Structural Elasticity. With piston-theory forcing and temperature-dependent natural frequency, per mode i our modal equations are:

$$\ddot{\eta}_i + 2\zeta_i \omega_i(T) \dot{\eta}_i + \omega_i(T)^2 \eta_i = Q_i(x, \eta, \dot{\eta}). \quad (\text{A.9})$$

A.2 Polynomial Embedding via Adjunction

A priori, the dynamics contain transcendental and rational terms. We repair each of these by adjoining one variable and one polynomial defining constraint, so the entire augmented system is polynomial in the augmented state:

1. **Transcendental Incidence:** $\alpha = \text{atan2}(w, u)$, $\beta = \arcsin(v/V)$ are inverse-trigonometric functions of body rates, we replace them with polynomial direction-cosine ratios $(u/V, v/V, w/V)$.
2. **Square-Root Speed:** Since $V = \sqrt{u^2 + v^2 + w^2}$ carries a square root, we adjoin V with the velocity magnitude constraint $V^2 = u^2 + v^2 + w^2$.
3. **Exponential Atmosphere:** Since $\rho = \rho_0 e^{-h/H_s}$ with $h = \|\vec{r}\| - R_e$ appears in $\bar{q} = \frac{1}{2} \rho V^2$, we adjoin $y = \sqrt{\rho/\rho_0}$ with rate $\dot{y} = -\frac{y}{2H_s} \frac{d}{dt} \|\vec{r}\|$, where the radial rate $\frac{d}{dt} \|\vec{r}\| = \varrho(\mathbf{R}\vec{r}) \cdot \vec{V}$ is polynomial once ϱ (item 4) is adjoined.

4. **Reciprocal Radius:** Gravity carries $\|\vec{r}\|^{-3}$, $\|\vec{r}\|^{-5}$, $\|\vec{r}\|^{-7}$, so we adjoin $\varrho = 1/\|\vec{r}\|$ with $\varrho^2 \|\vec{r}\|^2 = 1$, $\varrho > 0$, and rate $\dot{\varrho} = -\varrho^3 (\mathbf{R}\vec{r}) \cdot \vec{V}$. Gravity is then polynomial:

$$\vec{g} = -\mu\varrho^3 \left[\left(1 + \frac{3}{2}J_2R_e^2\varrho^2 (1 - 5r_3^2\varrho^2)\right) \vec{r} + 3J_2R_e^2\varrho^2 r_3 \hat{z}_{\mathcal{E}} \right]. \quad (\text{A.10})$$

No trigonometric adjunctions are needed: no angle is a state, and latitude dependence enters only through $r_3\varrho = \sin \phi$.

5. **Quaternion Norm:** $|\vec{q}|^2 = 1$ is already a polynomial invariant (skew-symmetry of Ω preserves it), no adjunction necessary.

6. **Heat / Load-Factor Square Roots:** $\dot{Q} \propto yV^3$ is already polynomial; however, $n = \|\vec{F}_a\|/(mg_0)$ requires squaring to $(\bar{q}S)^2(C_X^2 + C_Y^2 + C_Z^2) \leq (n_{\max}mg_0)^2$.

The Augmented State. Collecting all coordinates:

$$\tilde{x} = \left(\underbrace{r_1, r_2, r_3, u, v, w, q_0, q_1, q_2, q_3, p, q, r_b}_{13, \text{ core}}, \underbrace{y, V, \varrho, Q_{\text{tot}}, m, s, \chi}_{7, \text{ adjointed}}, \underbrace{\eta, \dot{\eta}}_{2n_{\eta}, \text{ modal}} \right). \quad (\text{A.11})$$

References